

METHOD OVERLOADING AND METHOD OVERRIDING**Aim:**

To understand and implement method overloading and method overriding.

PRE LAB EXERCISE**QUESTIONS**

1.What is method overloading?

- **Method overloading** means having **multiple methods with the same name** in the **same class**, but with **different parameters** (number, type, or order).
- It is an example of **compile-time polymorphism**.

2.What is method overriding?

- **Method overriding** occurs when a **subclass provides a specific implementation** of a method already defined in its **parent class**, with the **same method signature**.
- It is an example of **runtime polymorphism**.

3.Difference between overloading and overriding.

Feature	Method Overloading	Method Overriding
Occurs in	Same class	Parent–child classes
Method name	Same	Same
Parameters	Must be different	Must be same
Return type	Can be same or different	Must be same (or covariant)
Polymorphism	Compile-time	Runtime
Inheritance	Not required	Required

IN LAB EXERCISE

Objective:

To demonstrate compile-time and runtime polymorphism.

PROGRAMS:

1.Student Result System (Method Overriding)

Description:

- Base class Student has method displayResult().
- Subclasses UGStudent and PGStudent override the method to show different grading systems.

Code :

```
import java.util.Scanner;
```

```
// Base class
```

```
class Student {
```

```
    String name;
```

```
    void displayResult() {
```

```
        System.out.println("Student Result");
```

```
    }
```

```
}
```

```
// UG Student subclass
```

```
class UGStudent extends Student {
```

```
    int marks;
```

```
    UGStudent(String n, int m) {
```

```
        name = n;
```

```
        marks = m;
```

```
    }
```

```
@Override
```

```
void displayResult() {  
    double percentage = (marks / 100.0) * 100;  
    System.out.println("UG Student: " + name);  
    System.out.println("Marks: " + marks);  
    System.out.println("Percentage: " + percentage + "%");  
}  
}
```

// PG Student subclass

```
class PGStudent extends Student {  
    double gpa;  
  
    PGStudent(String n, double g) {  
        name = n;  
        gpa = g;  
    }  
}
```

@Override

```
void displayResult() {  
    System.out.println("PG Student: " + name);  
    System.out.println("GPA: " + gpa + " / 10");  
}  
}
```

// Main class

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        // Input for UG student  
        System.out.print("Enter UG Student Name: ");
```

```
String ugName = sc.nextLine();
System.out.print("Enter UG Student Marks (out of 100): ");
int ugMarks = sc.nextInt();
sc.nextLine(); // consume newline

// Input for PG student
System.out.print("Enter PG Student Name: ");
String pgName = sc.nextLine();
System.out.print("Enter PG Student GPA (0-10): ");
double pgGpa = sc.nextDouble();

// Create objects
Student s1 = new UGStudent(ugName, ugMarks);
Student s2 = new PGStudent(pgName, pgGpa);

System.out.println("\n--- Student Results ---");
s1.displayResult();
System.out.println();
s2.displayResult();

sc.close();
}
}
```

OUTPUT:

```
<terminated> Class1 [Java Application] C:\Users\SWETHA\p2\pool\plugins\org.eclipse.justj.
```

```
Enter UG Student Name: swetha
Enter UG Student Marks (out of 100): 95
Enter PG Student Name: Asmitha
Enter PG Student GPA (0-10): 9
```

```
--- Student Results ---
```

```
UG Student: swetha
```

```
Marks: 95
```

```
Percentage: 95.0%
```

```
PG Student: Asmitha
```

```
GPA: 9.0 / 10
```

2. Calculator Program (Method Overloading)

Description:

Create a Calculator class with multiple add() methods to calculate:

- Addition of 2 integers
- Addition of 3 integers
- Addition of 2 double numbers

Code:

```
import java.util.Scanner;
```

```
class Calculator {
```

```
    int add(int a, int b) {
```

```
        return a + b;
```

```
    }
```

```
    int add(int a, int b, int c) {
```

```
        return a + b + c;
```

```
    }
```

```
    double add(double a, double b) {
```

```
        return a + b;
```

```
    }
```

```
}  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        Calculator calc = new Calculator();  
  
        System.out.print("Enter two integers: ");  
        int x = sc.nextInt();  
        int y = sc.nextInt();  
        System.out.println("Sum of two integers: " + calc.add(x, y));  
  
        System.out.print("Enter three integers: ");  
        int p = sc.nextInt();  
        int q = sc.nextInt();  
        int r = sc.nextInt();  
        System.out.println("Sum of three integers: " + calc.add(p, q, r));  
  
        System.out.print("Enter two decimal numbers: ");  
        double a = sc.nextDouble();  
        double b = sc.nextDouble();  
        System.out.println("Sum of two doubles: " + calc.add(a, b));  
  
        sc.close();  
    }  
}
```

Output:

Problems Javadoc Declaration Console Eclipse IDE for Java Developers
<terminated> Class1 [Java Application] C:\Users\SWETHA\.p2\pool\plugins\org.eclipse.justj.openj
Enter two integers: 9 9
Sum of two integers: 18
Enter three integers: 9 7 92
Sum of three integers: 108
Enter two decimal numbers: 0.9 0.9
Sum of two doubles: 1.8

POST LAB EXERCISE

1. Is return type important in method overloading and method overriding?

- **Method Overloading:**

Return type **alone is NOT important**

Overloading depends on **parameter list**

- **Method Overriding:**

Return type **must be same or covariant** (subclass type)

2. Can you overload a method by changing only the return type?

No, changing only the return type **does not** overload a method.

Why?

Java identifies methods using **method signature** (name + parameters), not return type.

```
int show() { }
```

```
double show() { }
```

3. Can static methods be overridden? Can they be overloaded?

- **Static methods cannot be overridden**

- They **can be overloaded**

- ◆ Static methods belong to the **class**, not to objects

- ◆ Method hiding happens, not overriding

```
static void display(int a) { }
```

```
static void display(int a, int b) { }
```

4.Can a method be overridden if the parameter list is different?

No, a method **cannot be overridden** if the parameter list is different.

✓ Same parameter list → Overriding

✓ Different parameter list → Overloading

Result:

Thus the method overloading and overriding concepts were implemented and executed successfully.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		